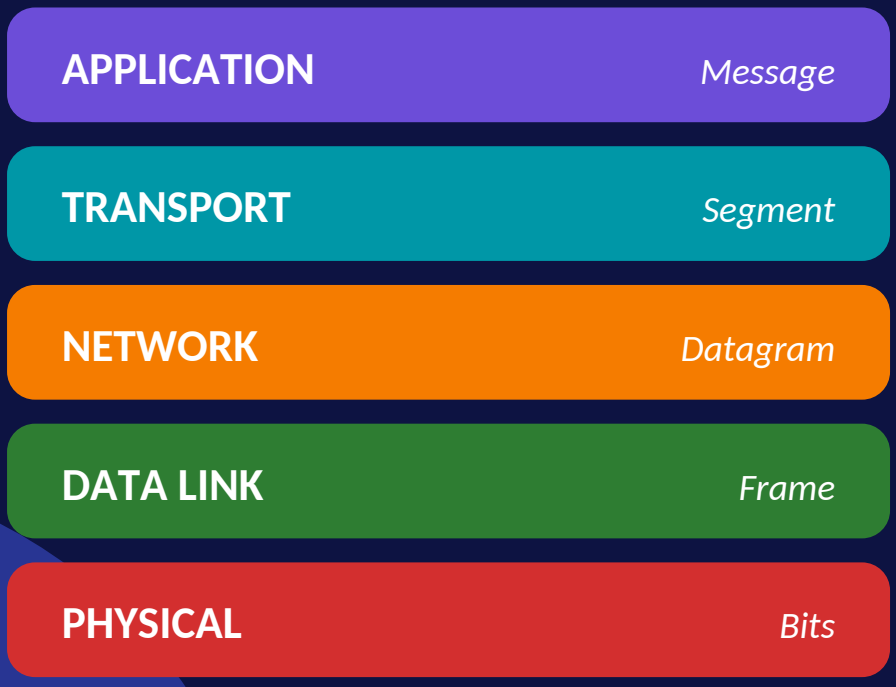
GATE 2026 | CRASH COURSE | LECTURE 1

Computer Networks

Switching, TCP/IP, OSI aur Layered Architecture

Easy Hinglish notes + colourful diagrams



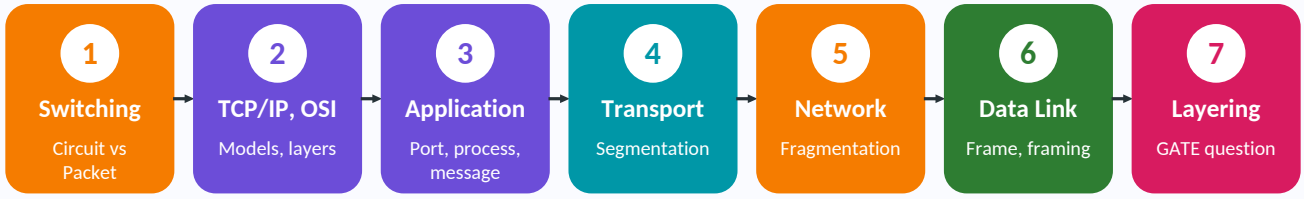
Is PDF me kya hai:

- Circuit vs Packet Switching
- TCP/IP aur OSI model ki poori kahani
- Har layer ka PDU, ports, protocols
- Segmentation, Fragmentation, Framing
- Worked GATE question + Quick revision sheet

Lecture ka Roadmap

Aaj kya-kya padhenge aur kyun

Ye pehli class hai, isliye **poore subject ka foundation** aaj hi bana rahe hain. Aaj ka goal simple hai: ek packet **kaise** source se destination tak jaata hai aur har layer par uska **naam** kya hota hai. Ye samajh gaya to aage ke saare chapters easy lagenge.



Ek hi lecture me: sab kuch top-down (Application se Physical tak)

Syllabus me kya likha hai?

- **Concept of layering** - OSI aur TCP/IP protocol stack
- **Basics of packet, circuit and virtual circuit switching** (virtual circuit is course me cover nahi hoga, time kam hai)

GATE TIP

Teacher ki advice: **syllabus ki har line zaroor dekho**. Syllabus me jo likha hai wo faltu jagah nahi gher raha. Switching par last question **GATE 2004** me aaya tha, par jis din paper-setter ne syllabus dhyan se dekha, pehla question yahin se banega. Intro wale topics se 1-mark question lagbhag har 2-3 saal me aa hi jaata hai.

Ye sab kahan se aaya? (Source of truth)

Books bhi **RFC (Request For Comments)** dekh kar likhi jaati hain. Computer Networks me standard research papers RFC naam se index ke saath milte hain. Kabhi book me confusion ho to RFC original source hai. GATE tak itna time nahi hai, isliye abhi bas crash course pe focus karo; RFC M.Tech me kaam aayenge.

Crash course ka plan (top-down)

Din	Topic	Kitne din / note
Kal	Application layer (DNS, HTTP, Email)	1 din, one-shot. Numericals bhi aate hain
Uske baad	Transport layer (TCP/UDP)	~3 din
Phir	Network layer	Sabse zyada weightage, sabse zyada din
Last	Data Link layer (CRC ka light version)	Last me, kyunki hum top-down chal rahe hain

YAAD RAKHO

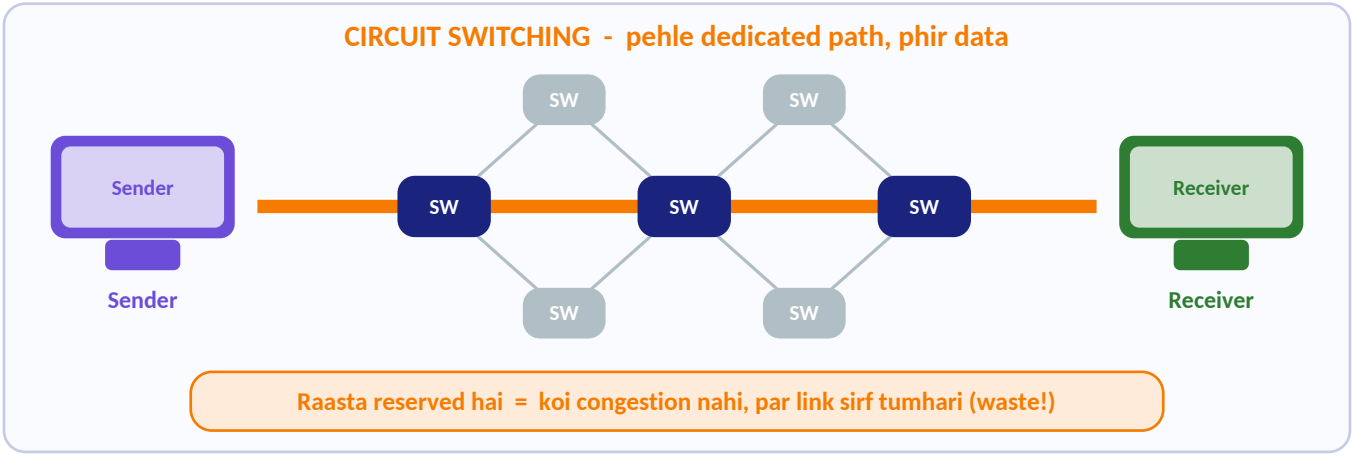
GATE exam ke din **3 ghante** ek hi chair par baithna hai. Washroom ke 5 minute = paper me 5 minute kam. Isliye padhai me bhi baithne ki aadat daalo (aur exam wale din paani kam piyo).

1 Switching: Circuit vs Packet

Data source se destination tak kaise move karta hai

Switching = data ko network ke through destination ki taraf move karne ka process. Iske **2 tareeke** hain: **Circuit Switching** (purana telephone network) aur **Packet Switching** (Internet).

A) Circuit Switching



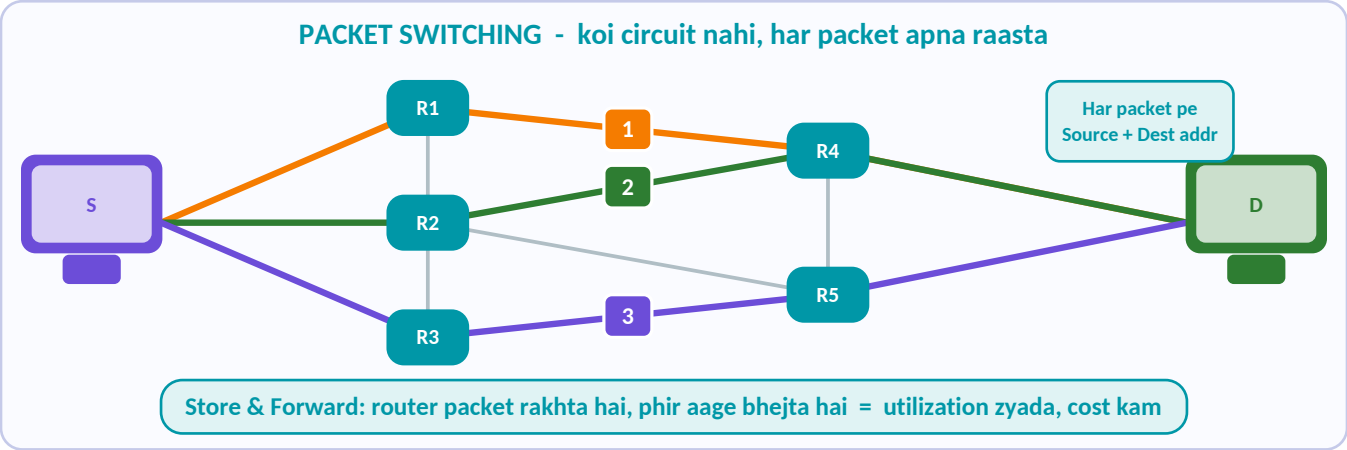
- Data bhejne se **pehle** sender aur receiver ke beech ek **dedicated (reserved) path** banana padta hai. Isse **path reservation / circuit establishment** kehte hain.
- Ye path **circuit switches** ki links se banta hai. Jab tak path nahi bana, **data transmission shuru hi nahi hota**.
- **Example:** Telephone network (landline, PSTN). Phone se operator, operator se operator, aur phir doosre user tak ek dedicated raasta.

Advantage (fayda)	Disadvantage (nuksaan)
Data transfer ke time koi congestion nahi. Ek baar line mil gayi to buffering / rukawat kabhi nahi, kyunki raasta sirf tumhare liye reserved hai.	1) Circuit banate time saari lines busy mil sakti hain ("all lines of this route are busy"). 2) Line **lease / rent** pe hoti hai, isliye per-minute / per-second charge lagta hai. 3) **Asli problem: inefficient utilization of network resources.**

EXAMPLE / ANALOGY

4 seater car me **akela insaan** office jaa raha hai, 3 seat khaali. Agar 4 log car-pooling karte to 4 alag cars ki zaroorat nahi padti. Circuit switching = akela car (resource waste). Packet switching = car-pooling / sharing bus.

B) Packet Switching



- Circuit switching ki **utilization problem** hatane ke liye US ki defence research agency **DARPA** (baad me **ARPA**) ne packet switching banayi. Pehle **network (sadak)** bana, phir uspe chalne wala **protocol / model (car) = TCP/IP**.
- **Koi circuit establish nahi karna padta.** Data ko **packets** me todte hain; har packet pe **source address aur destination address** likha hota hai. Routers inhe aage badhate hain. Isko **Store-and-Forward network** kehte hain.
- Source aur destination ke beech **koi dedicated path nahi** chahiye. **Example: Internet** (sabse bada WAN). Koi bank bhi leased lines pe apna packet-switched WAN bana sakta hai.

Advantage	Disadvantage
Resources **shared** mode me chalte hain, isliye **bandwidth utilization bahut better**. Ek hi time me **zyada users** support hote hain aur **cost kam** hoti hai.	Packets ke routing ke time **congestion** ho sakta hai (run-time condition). Isi wajah se video me **buffering** aati hai. Network fast hoga (high bandwidth) to congestion kam hota jaayega.

Side-by-side comparison

Point	Circuit Switching	Packet Switching
Path	Dedicated, pehle se reserved	Koi reservation nahi, shared
Setup	Data se pehle circuit establish	Zaroorat nahi
Congestion	Establish karte time (data ke time nahi)	Data transfer ke time (routing me)
Utilization	Poor (limited users)	Good (zyada users)
Charge / cost	Reserved line ka rent (per min/sec)	Kam
Example	Telephone (PSTN)	Internet
Train analogy	**Reserved coach** (1 berth = 1 insaan)	**Unreserved coach** (jo chadh sake chadh jaaye)

2 TCP/IP aur OSI Model

Layered architecture ki kahani

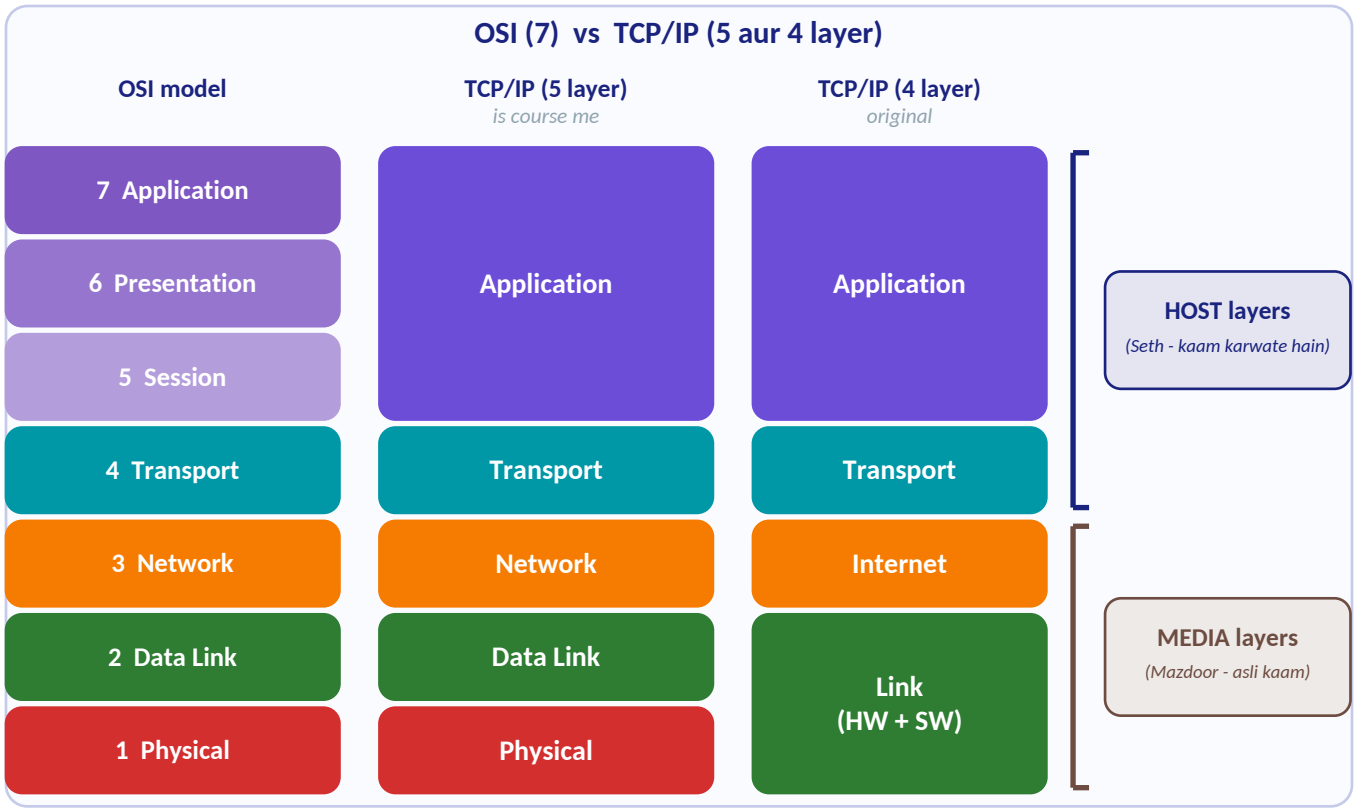
Pehle kaun aaya? (History)

- **Sabse pehle TCP/IP bana** (DARPA ne, lagbhag 1970 ya usse pehle). Pehle packet-switched network (sadak), phir uspe communication model TCP/IP (car).
- Fir gaon me ek ka ghar bana to doosre ko aag lagi: **IBM, Xerox, DEC** sabne apna-apna model bana liya. Confusion! Tab **ISO** aaya aur bola: ek **super-set** model banate hain.
- Isse bana **OSI = Open System Interconnection**. Ye ek **conceptual (theoretical)** model hai. Isme ek problem ke **multiple solutions** diye gaye hain, jo pasand aaye wo choose karo.
- OSI ne packet-switch network ko naya naam diya: **Datagram Network**. Naam badalne se kaam nahi badalta (chamgadad ka naam mor rakhne se wo mor nahi ban jaata).

DHYAN SE
Bahut log sochte hain OSI pehle bana aur usse TCP/IP kata gaya. **Ye galat hai**. Pehle TCP/IP bana, OSI baad me.

TCP/IP = Internet Protocol Stack (ya Suite)

- Ye ek **theoretical / conceptual model** hai: functions kitabon me likhe hain. Jab inko **code** karke software banate hain tab ye operational (implementation) model banta hai.
- Aapke computer ke **OS** ka communication architecture aur **router** ke andar ki programming, dono isi par based hain.
- **4 layer** (original) ya **5 layer** (Link layer me se hardware wala **Physical** alag kar diya). **Ye course 5-layer ke hisaab se chalega.** Layer ginti pe behas mat karna; examiner functionality puchta hai, number nahi.



Layer	OSI no.	TCP/IP me naam	Short role
Application	L7	Application	User jis application se interact karta hai (browser). Protocols: HTTP, DNS, SMTP
Presentation	L6	(Application me merge)	Code conversion (ASCII / non-ASCII). TCP/IP me **optional**
Session	L5	(Application me merge)	Session-based communication. TCP/IP me **optional**
Transport	L4	Transport	Process-to-process delivery (TCP, UDP)
Network	L3	**Internet** layer	Routing, IP. **Sirf naam badla, kaam same**
Data Link	L2	Link	Ek link par frame delivery
Physical	L1	Link (HW)	Channel me bits read / write / transmit

PYQ
ISRO ne pucha hai: upar ki 4 layers (Application..Transport) = **Host layers (Seth)**. Neeche ki 3 layers (Network, Data Link, Physical) = **Media layers (Mazdoor)**. Seth khud kaam nahi karte, mazdoor se karwate hain. Bas isi se yaad rakho.

YAAD RAKHO
Kaun kahan? User = Application layer (browser) se interact karta hai. **OS** = Presentation se Data Link tak. **Hardware** = Physical layer. Physical layer hi channel me bits likhta / padhta aur transmit / receive karta hai. OSI wale question me seedha **layer number** puchhte hain (jaise Layer 3 = Network), isliye numbers yaad rakho.

3 Application Layer

Process, port number, protocol aur message

Normal process vs Network process

- **Program** = compiler ka output / executable code, jo secondary memory me padha hota hai. **Process** = wo program jo **execution me** hai (main memory me load).
- Computer me hazaron processes hote hain, par **har process ko network nahi chahiye**. Jo process network communication karti hai usse **Network Process / Network Application Program** kehte hain.
- **Example:** Browser, WhatsApp = network process (internet chahiye). Calculator, Alarm clock = normal process (internet nahi chahiye). (Ye concept KBC me bhi aa chuka hai!)

Port Number: process ki ID

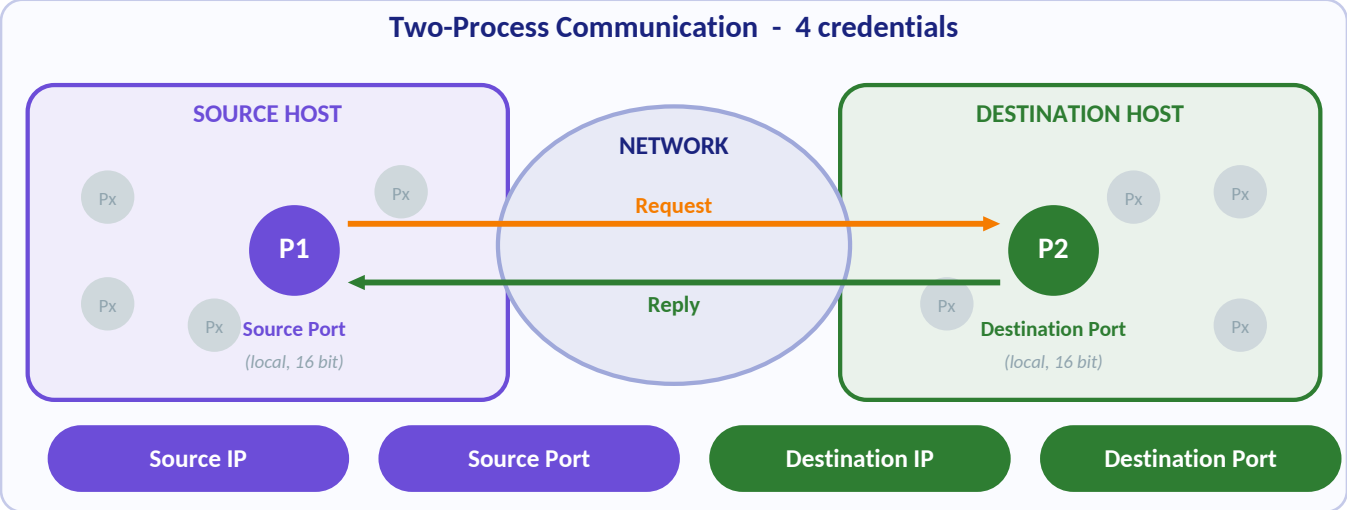
- Network application ko **port number** se identify karte hain. Ye **16 bit** ka hota hai = **2 byte** (1 byte = 8 bit). Ye network ka standard hai, hamesha 2 byte hi rahega.
- Port number **locally unique** hota hai (ek machine ke andar), **globally unique nahi**. Isliye ise **local port number** bhi kehte hain.
- **Process ID kaafi kyun nahi?** OS har process ko PID deta hai, par port number sirf unhe milta hai jinhe network chahiye.

EXAMPLE / ANALOGY

Aadhar card sabke paas hota hai (= Process ID, har process ke paas). **Passport** sirf unke paas jinhe videsh jaana hai (= Port number, sirf network processes ke paas).

Browser ke tabs: do tabs me google.com khola, ek me "CN books", doosre me "DBMS books". IP address same hai, to reply sahi tab me kaise aayega? **Alag port number se.**

Two-Process Communication: 4 chizein chahiye



Communication **hosts** ke beech nahi, **processes** ke beech hoti hai (ghar baat nahi karte, log karte hain). P1 (source) -> P2 (destination) ko message bhejti hai. Destination host **destination IP** se milta hai, aur uske andar P2 **destination port** se. P2 ko ye bhi jaanna hota hai ki data kaun bhej raha hai, isliye **source IP + source port** bhi jaata hai (jaise letter pe apna address bhi likhte ho). Reply me roles ulat jaate hain.

YAAD RAKHO

Do network processes **hamesha alag-alag hosts** pe hote hain aur **messages exchange** karke network par communicate karte hain.

Protocol, client-server aur PDU

- **Protocol = set of rules**: data kaise exchange hoga (jaise car chalane ka poora procedure hota hai).
- Alag server, alag protocol. **Jaisa desh, waisa bhesh**: client ko server ke hisaab se protocol badalna padta hai. Server ka protocol fixed, **client dal-badlu**.
- **Dono taraf same protocol** chalta hai (HTTP client <-> HTTP server). Cross-protocol nahi chalta (DNS ka data SMTP ko nahi jaata). Isse kehte hain *same protocol of different machines*.

Protocol	Server	Kab use hota hai
DNS	DNS server	Domain name se IP address pata karna
HTTP	Web server	Web resources / pages maangna
SMTP	Mail server	Mail bhejna

PDU (Protocol Data Unit) = ek baar me exchange hone wali basic unit. Application layer ka PDU hai **MESSAGE**: HTTP request message / response message, DNS request / response message, SMTP mail message. OSI ne iska nickname *"data / user data"* rakha, par **exam me "Message" likhna hai**, nickname nahi.

PYQ
GATE 2012 (IIT Delhi), 1 mark: *"PDU for the application layer in the Internet stack is?"* Answer: **Message** (lecture ke slide me ye option D tha). Isliye ye topic fizool nahi, ek baar aaya to dobara bhi aa sakta hai.

GATE TIP
GATE 2025 ke Set 1 aur Set 2 dono me Application layer se question **nahi** aaya. Teacher ke hisaab se is saal wapas aane ki umeed hai. Application layer me **numericals bhi** aate hain.

4 Transport Layer

TCP / UDP aur Segmentation

Do protocol: UDP aur TCP

	UDP (User Datagram Protocol)	TCP
Nature	**Light-weight**, minimal functionality, basic	**Pro / advanced** protocol
Services	Sirf basic services	UDP ki saari services **+ extra add-on services**
Reliability / QoS	Unreliable, QoS ki umeed nahi	**Reliable**, Quality of Service dene wala

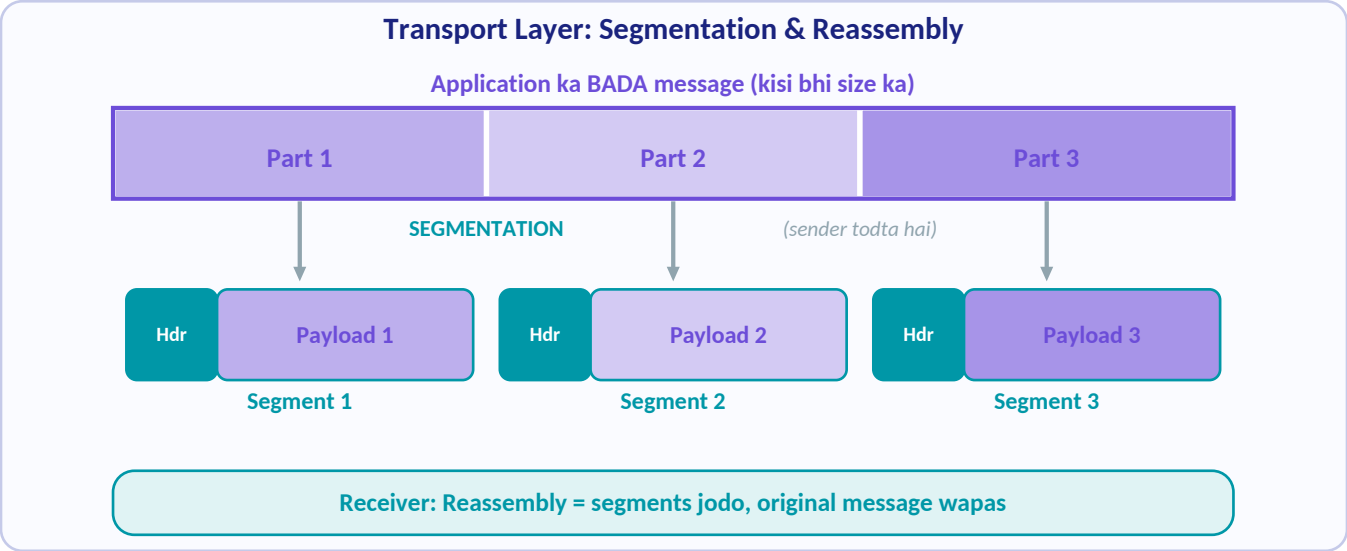
EXAMPLE / ANALOGY
Sui ka kaam sui karti hai, talwar ka kaam talwar. Jahan UDP ki zaroorat hai wahan TCP kaam nahi aata, aur jahan TCP chahiye wahan UDP. Dono ki apni jagah hai. Detail me TCP / UDP 2 din baad padhenge.

Application layer ka process decide karta hai ki use **TCP** chahiye ya **UDP**. Jisko choose kiya, usko apna message de deta hai.

Transport layer ka PDU = SEGMENT

- Message ko **payload** maan kar uske aage **TCP header** (ya **UDP header**) laga dete hain. Header me transport layer apni info likhta hai (sender / receiver ke liye), payload me upar wali layer ka data hota hai.
- **TCP** ka PDU = **TCP segment**. **UDP** ka PDU = pehle *UDP datagram*, **latest norm** me *UDP segment*. Dono naam chalte hain; GATE 2025 ke fragmentation question me "UDP segment" word aaya tha.

Segmentation aur Reassembly



- **Message ki size par koi restriction nahi** (kisi bhi size ka ho sakta hai). Restriction **segment** ki size par hai (kitni, ye aage ke chapter me). Ye GATE 2008 me puchha ja chuka hai.
- Message chhota hai to **ek payload** me aa jaata hai (1 segment). Bada hai to **multiple payloads** me toot kar **multiple segments** bante hain (3, 4, 5, jitne zaroori ho).
- **Segmentation** = message se segments banana (sender side). **Reassembly** = segments ko jod kar wapas message banana (receiver side).

EXAMPLE / ANALOGY
Library shift karni hai. Kitabein kam hain to ek pickup van kaafi. Zyada hain to **kai pickup vans** me baant ke bhejo aur doosri taraf phir jod do. Yaad rakho: **Sender todta hai, Receiver jodta hai.** (Hyderabad Ramoji Film City wale Bhallaldeva ke putle jaise: bada putla 3 hisson me, sar, dhad, pair.)

GATE TIP
1 mark ka question ban sakta hai: **"Message se segment banane ki process ko kya kehte hain?"** Answer: **Segmentation**.

5 Network Layer (Internet Layer)

IP, Datagram aur Fragmentation

Sirf ek hi protocol: IP

- Application layer me **bahut** protocols, Transport me **2** (TCP, UDP), lekin Network layer me **sirf ek: IP**. Iska koi alternate / substitute nahi hai. Sab IP packets isi protocol se move karte hain.
- **IPv4 vs IPv6:** version badalne se protocol alag nahi hota, functionality badalti hai. **IPv4 syllabus me hai**. IPv6 syllabus me 2016 se 2020 tak tha, **2021 me hata diya gaya**.

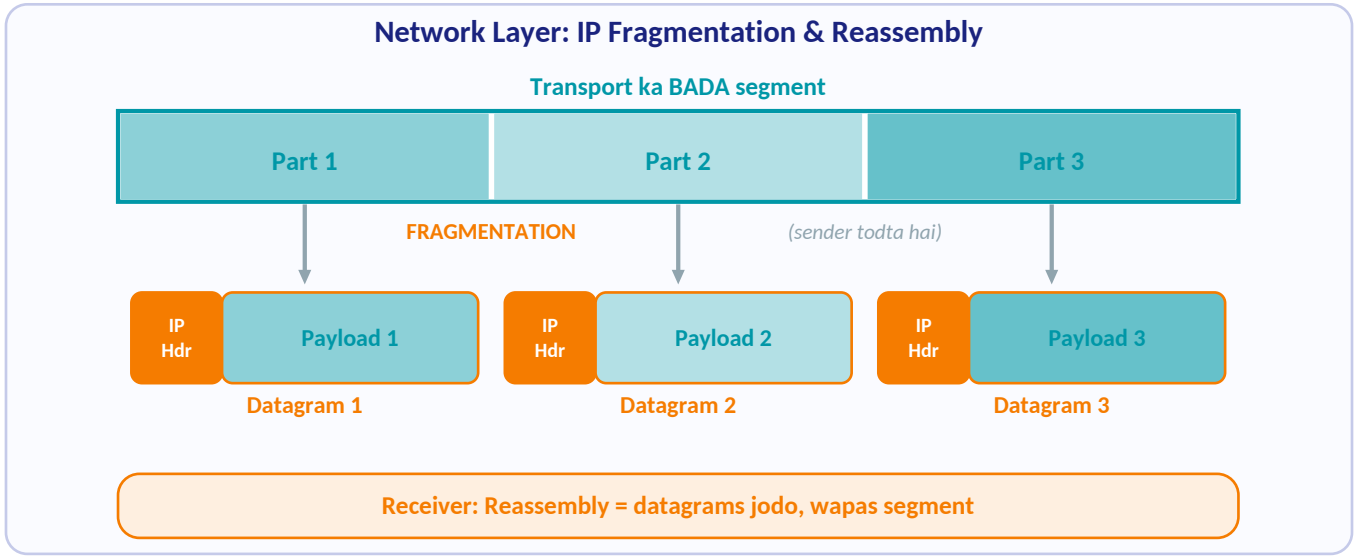
Network layer ka PDU = DATAGRAM

- TCP/IP isse **packet** bolta hai, OSI **datagram**. Kahin **IP datagram**, kahin **IP packet**, kahin **IP fragment** milega: **ek hi cheez ke 3 naam**. Exam me **Datagram** likho.
- Datagram = **IP header + payload**. Payload me transport layer ka **segment** hota hai.
- Network layer jab datagram banata hai, tab data link layer se pooch kar banata hai ki **kitne size ka** banana hai.

EXAMPLE / ANALOGY

Truck ke payload me ek **pickup van** rakhi, pickup van ke payload me ek **bike** (message). Har gaadi ka apna payload hota hai. Jaise neeche jaate ho, payload bada hota jaata hai kyunki wo upar ki layers ko **encapsulate** karta hai.

Fragmentation aur Reassembly



- Agar segment bada ho aur ek IP datagram ke payload me na aaye, to wo **multiple IP datagrams** me toota jaata hai. Is process ko **IP Fragmentation** kehte hain.
- Receiver side par ulta hota hai: **datagrams ko jod kar segment** banta hai (**reassembly**), aur transport layer saare segments jod kar message bana deti hai.

	Kaun todta hai	Kya banta hai	Process ka naam
Transport layer	Message -> Segments	Segment	**Segmentation**
Network layer	Segment -> Datagrams	Datagram	**Fragmentation**

YAAD RAKHO

Hamesha: **Sender todta hai, Receiver jodta hai.** Sender side jo bhi hua, receiver side uska **ulta (reverse)** hota hai.

6 Data Link aur Physical Layer

Frame, Framing aur Bits

Data Link layer ka PDU = FRAME

- Frame ke **3 parts** hote hain: **Header**, **Payload** (isme IP datagram) aur **Footer**. Footer sirf frame me hota hai (segment / datagram me nahi).
- **Footer me CRC / error-check bits** aati hain. CRC payload ke baad hi aati hai kyunki wo hardware se generate hoti hai. (Error control chapter me detail, last me.)
- Chahe TCP ho ya UDP, dono ne Data Link ke PDU ke liye **ek hi naam** use kiya: **Frame**. Koi choice nahi.

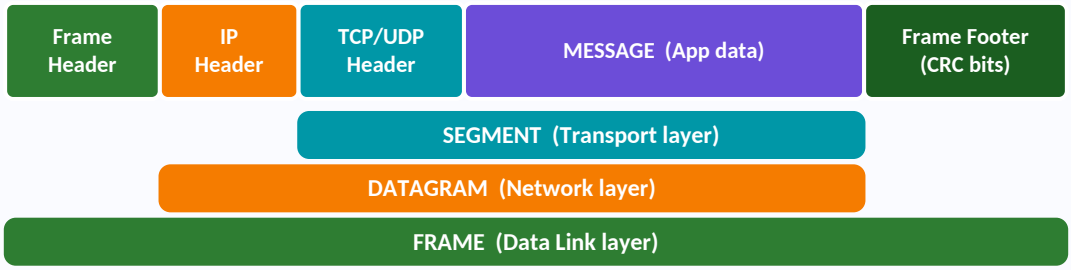
Ek IP datagram = ek frame

DHYAN SE

Network layer ko **bade IP datagram banane ki permission nahi** hai. Data Link layer bolta hai: "mere ek frame ke payload me jitna maximum aa sakta hai, utna hi banao, warna tum khud deliver karo." Isliye **IP datagram kabhi do frames me nahi tootta** aur **do datagrams ek frame me nahi jaate**. Ek datagram = ek frame.

Encapsulation - ek frame ke andar poori kahani

Sabse andar Message - upar se header lagte jaate hain (ek truck, uske andar pickup, uske andar bike)



Physical layer poore frame ko bit-by-bit channel par bhejta hai (PDU = Bits)

Process	Kahan	Kya hota hai
Framing	Sender	IP datagram ko **ek frame me encapsulate** karo, phir Physical layer ko do
Extraction	Receiver	Frame ke payload se **IP datagram nikaalo**, aur Network layer ko do

Physical layer ka PDU = BITS

Physical layer **frame ko bit-by-bit** channel par transmit karta hai (aur receiver side par bit-by-bit receive karke frame banata hai). Bits ko transmit karne ka kaam yahi karta hai. Baaki saari layers ek doosre ko kaam **bypass** karti hain; **asli mazdoor Physical layer hai**.

GATE TIP

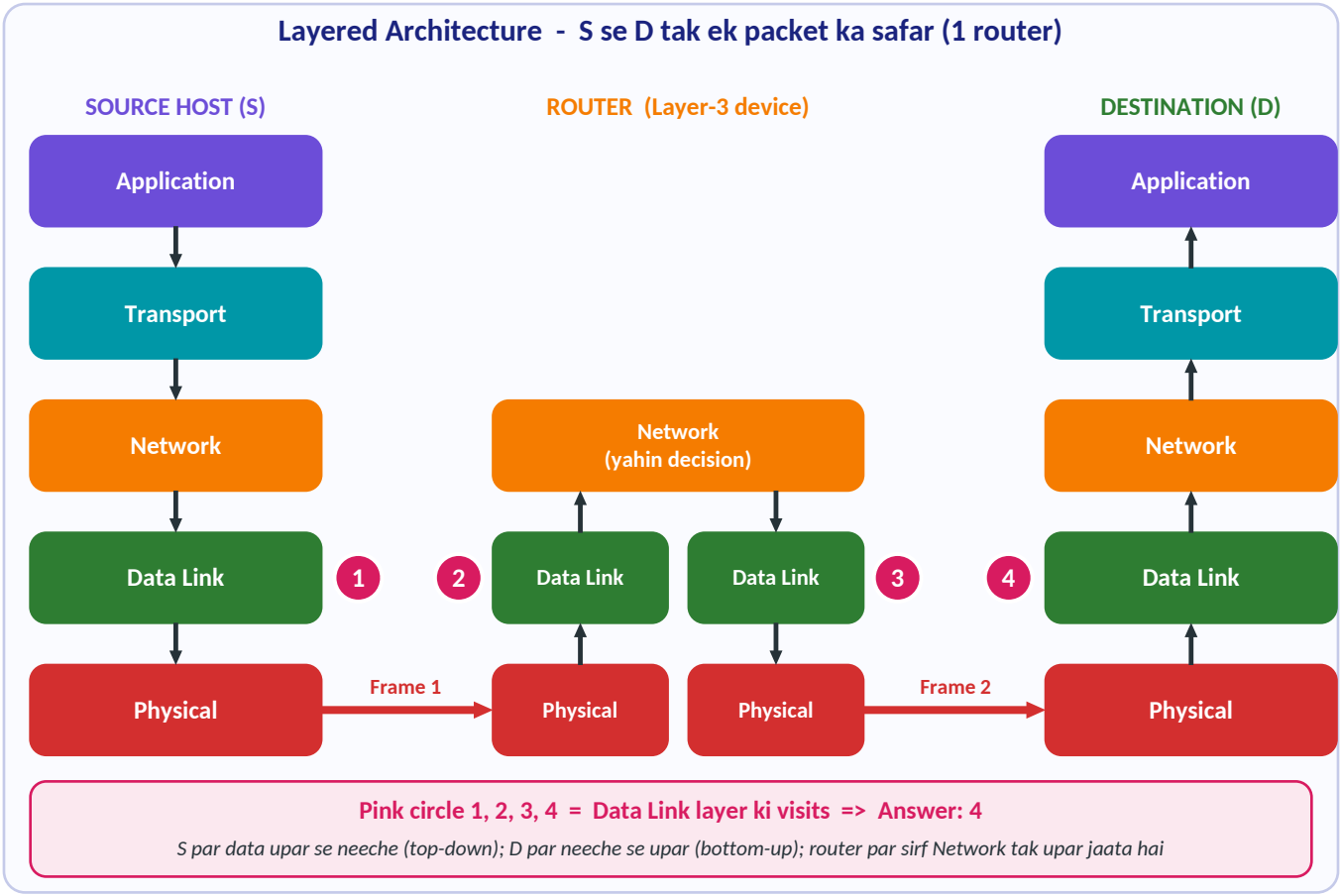
Chaaron bolo, ek saath: **Application = Messages**, **Transport = Segments**, **Network = Datagrams**, **Data Link = Frames**, **Physical = Bits** (ye short notes me zaroor likhna, 1-mark me kaam aayega).

7 Layered Architecture: Worked Example

Concept of layering ko ek question se samjho

PYQ

Question (class poll): Source host S aur destination host D **ek router** se connected hain. S se D tak ke transmission me ek packet **Data Link layer ko kitni baar visit** karega?



Step-by-step solution

- **Visit 1:** S ke andar message neeche aata hai: Application -> Transport (segment) -> Network (datagram) -> **Data Link (frame 1)** -> Physical.
- **Visit 2:** Frame 1 bits me router tak jaata hai. Router ki Physical -> **Data Link** frame kholta hai, datagram nikaalta hai -> Network layer.
- Router **Layer-3 device** hai, isliye **decision Network layer par** hota hai (kis taraf forward karna hai). Router ko pata chalta hai ki D mere se connected hai, aur wo us interface ki Data Link ko datagram deta hai.
- **Visit 3:** Router ki doosri taraf ki **Data Link** naya **frame 2** banati hai, Physical se bits bhejti hai.
- **Visit 4:** D ki Physical -> **Data Link** frame kholti hai -> datagram -> Network -> Transport -> Application (message P2 tak).
- **Answer = 4.** (Class me 66% ne 4 hi bataya.)

EXAMPLE / ANALOGY

Do station ke beech direct train nahi hai. Aap **IP datagram** ho, **Frame 1** pehli train, beech ke station (router) par utre, **Frame 2** doosri train pakdi, aur aage chal diye. **Train (frame) badli, yatri (datagram) wahi raha.**

YAAD RAKHO

Router ke **har interface (port)** ke liye alag Physical + Data Link ki pair hoti hai, lekin **Network layer ek hi** hoti hai. Router sirf **Layer 1, 2, 3** chalata hai. Agar beech me **n routers** hon to Data Link visits = **2n + 2** (n = 1 ke liye 4; ye pattern upar ke logic se nikalta hai).

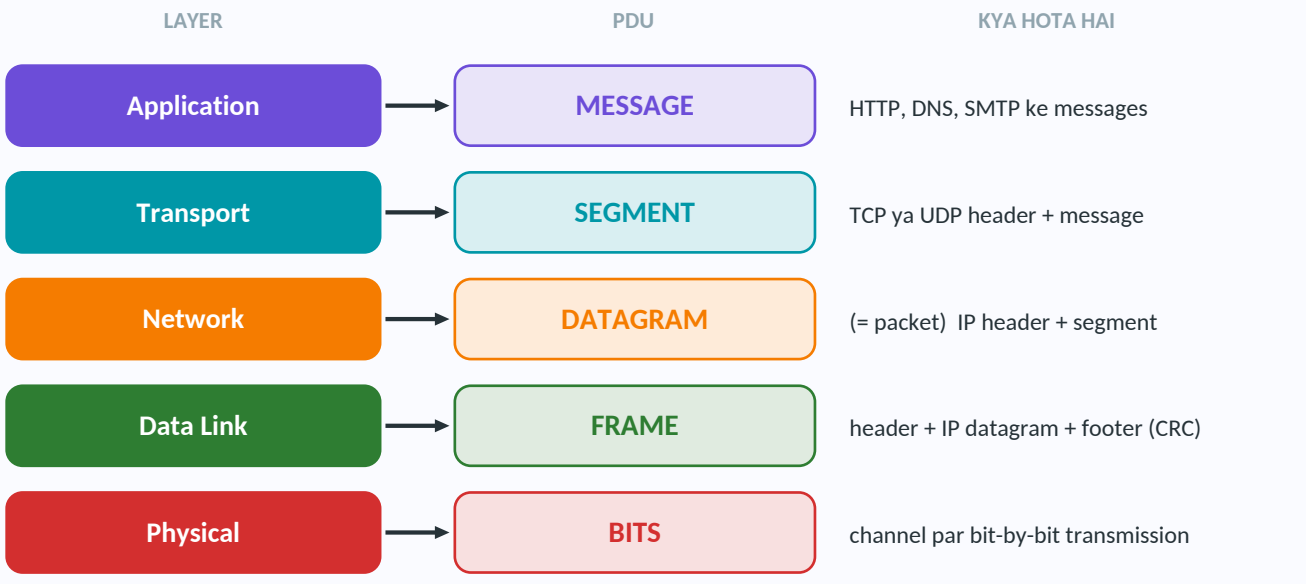
Concept of Layering ka saar

- Har layer apni **upar wali layer ko service** deti hai aur **niche ka sab kuch abstract** kar deti hai. Process P1 ko kabhi pata nahi chalta ki Transport layer ke baad data kahan gaya.
- Transport layer **processes** ko service deta hai. Network layer **Transport** ko, Data Link **Network** ko, Physical **Data Link** ko. Sab ek doosre ke order maan rahe hain, taaki upar wali layers ka kaam ho sake. Credit boss layer le jaati hai, asli mazdoori Physical karta hai.
- **Sender side:** data **upar se neeche** (top-down). **Receiver side: neeche se upar** (bottom-up).
- Subject padhne ke 2 tareeke: **Top-down** (Application se shuru, ye crash course) aur **Bottom-up** (Physical / Data Link se shuru, paid batches).

8 Quick Revision Sheet

Exam se pehle bas ye ek page dekh lo

Har layer ka PDU - ek nazar me



Layer	Naam / OSI no.	PDU	Host ya Media	Kaam
Application	L7	**Message**	Host (Seth)	User se interact, network process
Transport	L4	**Segment**	Host (Seth)	Process-to-process, TCP / UDP
Network	L3	**Datagram** (= packet)	Media (Mazdoor)	Routing, IP. Router = L3 device
Data Link	L2	**Frame**	Media (Mazdoor)	Header + payload + footer (CRC)
Physical	L1	**Bits**	Media (Mazdoor)	Bits channel par transmit

Yaad rakhne wali 12 lines

- Circuit switching = **dedicated path**, no congestion during transfer, par **poor utilization**. Packet switching = **shared**, better utilization, par congestion in routing.
- **Internet = packet switching** (store-and-forward), banaya **DARPA** ne. TCP/IP **OSI se pehle** aaya.
- TCP/IP = 4 ya 5 layers; OSI = 7 layers. Extra OSI layers = **Presentation, Session**. Network (OSI) = Internet (TCP/IP).
- Upar ki 4 = **Host layers (Seth)**. Neeche ki 3 = **Media layers (Mazdoor)**.
- **Port** = 16 bit = 2 byte, **local unique**, sirf network processes ke paas. PID sabke paas.
- Two-process communication ke liye **4 cheezein**: source IP, source port, dest IP, dest port.
- Protocol dono taraf **same**. Client server ke hisaab se protocol badalta hai.
- Message ki size pe **koi limit nahi**; segment aur datagram ki size pe limit hai.
- **Segmentation** (message -> segments), **Fragmentation** (segment -> datagrams), **Framing** (datagram -> frame).
- **Sender todta hai, Receiver jodta hai.**
- **Ek IP datagram = ek frame**, kabhi nahi tootta aur kabhi do saath nahi.
- **Router (L3)** ke case me packet Data Link ko **4 baar** visit karta hai (1 router).

Is lecture ke PYQ hints

Year	Topic	Kya aaya
2004	Switching	Question aaya tha, uske baad se nahi
2008	Message size	Message kisi bhi size ka ho sakta hai
2012 (IIT Delhi)	PDU	Application layer ka PDU = Message
2025	Fragmentation	Question me "UDP segment" word use hua
ISRO	Host vs Media layers	Upar ki 4 host layers
IIT Bombay (1 mark)	Layering / PDU	Homework question (Telegram channel pe discuss)

GATE TIP
Kal ka topic: **Application Layer** (DNS, HTTP, Email). Aaj ke PDU names ratt lo, kyunki aage har chapter me wahi kaam aayenge. **Homework:** IIT Bombay ka question khud try karo, phir typed solution se check karo.